

RAMANUJAN PERIOD SPECTRA AND RIGIDITY OF UNITARY CAYLEY SHIFTS

ANONYMOUS

ABSTRACT. For $n \geq 2$, let X_n be the nearest-neighbor shift on $\mathbb{Z}/n\mathbb{Z}$ in which x_j may be followed by x_{j+1} exactly when their difference is a unit modulo n . We compute its complete periodic spectrum from Ramanujan sums:

$$\#\text{Fix}(\sigma^k) = \sum_{d|n} \varphi(d) \left(\mu(d) \frac{\varphi(n)}{\varphi(d)} \right)^k.$$

Consequently

$$\zeta_{X_n}(z) = \prod_{\substack{d|n \\ \mu(d) \neq 0}} \left(1 - \mu(d) \frac{\varphi(n)}{\varphi(d)} z \right)^{-\varphi(d)}.$$

The shift is irreducible with entropy $\log \varphi(n)$; it is mixing for odd n and has exact period two for even n . Its unique maximal-entropy law is the uniform unit-increment Markov chain. For odd n , correlations of vertex observables decay at the sharp spectral rate $(p_{\min} - 1)^{-k}$, where $p_{\min}$ is the least prime factor of n . Finally, entropy together with the two-period count $n\varphi(n)$ recovers n , so distinct members of the family are not topologically conjugate; equivalently, the dynamical zeta function is rigid inside this family.

1. INTRODUCTION

The unitary Cayley graph on $\mathbb{Z}/n\mathbb{Z}$ joins two residues when their difference is invertible modulo n . Its graph spectrum is classical: Fourier characters diagonalize its adjacency matrix and the eigenvalues are Ramanujan sums. Klotz and Sander established this formula and a broad set of graph-theoretic properties [2]. We use that owned spectral input to solve the symbolic dynamics of the associated edge constraint.

For a finite-type shift with adjacency matrix A , periodic counts are $\text{tr}(A^k)$ and the Bowen–Lanford identity is $\zeta(z) = \det(I - zA)^{-1}$ [1]. Applying these standard facts to the Ramanujan spectrum produces more than a restatement of graph regularity: it gives a divisor-indexed orbit spectrum, an arithmetic mixing-period dichotomy, a sharp correlation rate, and a two-observable conjugacy classifier.

The narrow contribution is this complete dynamical package for the unitary Cayley shift. We do not claim the graph definition, its integral spectrum, Ramanujan-sum identities, the Parry construction, or the finite-type zeta determinant.

2. DEFINITION AND IRREDUCIBILITY

Let $U_n = (\mathbb{Z}/n\mathbb{Z})^\times$ and let A_n be the $n \times n$ zero-one matrix

$$(A_n)_{ab} = \mathbf{1}_{\{b-a \in U_n\}}, \quad a, b \in \mathbb{Z}/n\mathbb{Z}.$$

Define the two-sided vertex shift

$$X_n = \{x \in (\mathbb{Z}/n\mathbb{Z})^{\mathbb{Z}} : A_{x_j x_{j+1}} = 1 \text{ for every } j\},$$

with left shift σ . Every row and column of A_n has $\varphi(n)$ ones. Since $1 \in U_n$, the directed graph is connected.

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Proposition 2.1 (period and entropy). *For every $n \geq 2$, X_n is irreducible and*

$$h_{\text{top}}(X_n) = \log \varphi(n).$$

If n is odd, X_n is mixing. If n is even, its exact topological period is two.

Proof. Steps by $+1$ connect all residues, proving irreducibility. The constant vector is a positive eigenvector of A_n with eigenvalue $\varphi(n)$; the maximum row sum bounds the spectral radius from above by the same number. The entropy formula follows.

Every edge can be immediately reversed because $-U_n = U_n$, so closed walks of length two exist. If n is odd, n successive $+1$ steps give an odd closed walk; the gcd of 2 and n is one. If n is even, every unit is odd, so every step changes parity and all closed walks have even length. The existing two-cycles make the period exactly two. $\square$

3. RAMANUJAN ORBIT SPECTRUM

Let $\omega = e^{2\pi i/n}$. The Fourier vector $v_r = (\omega^{ra})_{a \in \mathbb{Z}/n\mathbb{Z}}$ is an eigenvector of A_n with eigenvalue

$$c_n(r) = \sum_{u \in U_n} \omega^{ru},$$

the Ramanujan sum. If

$$d = \frac{n}{\gcd(n, r)},$$

then the classical formula, in the notation convenient here, is

$$(1) \quad c_n(r) = \mu(d) \frac{\varphi(n)}{\varphi(d)}.$$

Exactly $\varphi(d)$ residues r give a specified divisor $d \mid n$. Indeed, they are $r = (n/d)s$ with s running through the reduced residue classes modulo d . This includes the unique class $r = 0$ for $d = 1$.

Theorem 3.1 (periodic points and zeta). *For every $n \geq 2$ and $k \geq 1$,*

$$(2) \quad P_k(n) := \# \text{Fix}_{X_n}(\sigma^k) = \sum_{d \mid n} \varphi(d) \left(\mu(d) \frac{\varphi(n)}{\varphi(d)} \right)^k.$$

The Artin–Mazur zeta function is the rational function

$$(3) \quad \zeta_{X_n}(z) = \prod_{\substack{d \mid n \\ \mu(d) \neq 0}} \left(1 - \mu(d) \frac{\varphi(n)}{\varphi(d)} z \right)^{-\varphi(d)}.$$

In particular,

$$P_1(n) = 0, \quad P_2(n) = n\varphi(n).$$

Proof. A point fixed by σ^k is a based closed walk of length k , so its number is $\text{tr}(A_n^k)$. Summing the k th powers of the eigenvalues in (1), grouped by d , proves (2). The determinant identity of Bowen and Lanford gives

$$\zeta_{X_n}(z) = \det(I - zA_n)^{-1};$$

inserting the eigenvalues and omitting zero factors gives (3). There are no self-loops, proving $P_1 = 0$. Alternatively, a based two-step loop is specified by its starting residue and one of $\varphi(n)$ unit increments, so $P_2 = n\varphi(n)$. $\square$

For example, if p is prime then X_p is the complete graph shift without self-loops and

$$\zeta_{X_p}(z) = \frac{1}{(1 - (p-1)z)(1+z)^{p-1}}.$$

If $n = 2^a$, the only nonzero adjacency eigenvalues are $\pm 2^{a-1}$, each simple, and

$$\zeta_{X_{2^a}}(z) = \frac{1}{1 - 2^{2a-2}z^2}.$$

The remaining $n - 2$ zero eigenvalues are invisible to periodic counts but not to the alphabet size; the two-period formula retains that size.

4. THE MAXIMAL MEASURE AND A SHARP MIXING RATE

Because A_n is irreducible, the shift has a unique measure of maximal entropy, the Parry measure [3]. Regularity makes it elementary: the stationary distribution is uniform on $\mathbb{Z}/n\mathbb{Z}$ and

$$(4) \quad \mathbb{P}(x_{j+1} = b \mid x_j = a) = \frac{(A_n)_{ab}}{\varphi(n)}.$$

Equivalently, the increments $x_{j+1} - x_j$ are independent and uniform on U_n , with an independent uniform initial residue.

For odd n , let $p_{\min}$ be its smallest prime divisor. The normalized transition operator $K_n = A_n/\varphi(n)$ is self-adjoint in uniform L^2 , and (1) gives its eigenvalues

$$(5) \quad \frac{c_n(r)}{\varphi(n)} = \frac{\mu(d)}{\varphi(d)}.$$

Among $d \mid n$, $d > 1$, the largest nonzero modulus is $1/(p_{\min} - 1)$.

Theorem 4.1 (sharp correlation rate). *Suppose n is odd. For mean-zero real-valued functions f, g on $\mathbb{Z}/n\mathbb{Z}$,*

$$|\mathbb{E}_{\mu_n}[f(x_0)g(x_k)]| \leq (p_{\min} - 1)^{-k} \|f\|_{L^2(u_n)} \|g\|_{L^2(u_n)},$$

where u_n is the uniform law. The exponential factor is sharp for every k : equality is attained by normalized real eigenfunctions whose associated divisor is $p_{\min}$.

Proof. On the orthogonal complement of constants, (5) bounds the operator norm of K_n^k by $(p_{\min} - 1)^{-k}$. The Markov correlation is $\langle f, K_n^k g \rangle_{L^2(u_n)}$, so Cauchy–Schwarz gives the inequality. The divisor $p_{\min}$ occurs with multiplicity $\varphi(p_{\min})$ and has normalized eigenvalue $-1/(p_{\min} - 1)$. The real or imaginary part of a nonzero Fourier eigenvector is a real eigenfunction; taking $f = g$ to be its normalization proves sharpness after the absolute value. $\square$

For even n , the eigenvalue -1 arising from $d = 2$ is the spectral form of the parity obstruction in [proposition 2.1](#); ordinary mixing cannot hold.

5. RIGIDITY INSIDE THE FAMILY

Euler’s totient is not injective, so entropy alone does not classify these shifts. The first nonzero periodic statistic repairs exactly that loss.

Theorem 5.1 (two-statistic conjugacy rigidity). *If X_m and X_n , with $m, n \geq 2$, are topologically conjugate, then $m = n$. More precisely, the pair*

$$\left(h_{\text{top}}(X_n), \# \text{Fix}(\sigma^2) \right) = (\log \varphi(n), n\varphi(n))$$

recovers n . Hence equality of the dynamical zeta functions also forces $m = n$ within this family.

Proof. Conjugacy preserves entropy and periodic-point counts. Equality of the entropies gives $\varphi(m) = \varphi(n)$. Equality of the two-period counts then gives $m\varphi(m) = n\varphi(n)$, hence $m = n$. If the zeta functions are equal, all periodic counts agree. Their exponential growth recovers the entropy of an irreducible finite-type shift. More precisely, Perron–Frobenius theory gives

$$\limsup_{k \rightarrow \infty} P_k(n)^{1/k} = \varphi(n),$$

so the same argument applies even in the period-two case. $\square$

6. CONTROLS AND OWNERSHIP BOUNDARY

The deterministic control constructs A_n directly, multiplies it over the integers, compares traces with (2), checks the grouped Ramanujan characteristic polynomial, verifies parity/mixing witnesses, and tests the rigidity pair over a finite registry. These computations guard the divisor indexing and endpoints; the proofs establish the formulas for all n .

The adjacency spectrum, including its Ramanujan-sum expression, belongs to the unitary-Cayley-graph literature [2]. Rational zeta and trace identities for finite-type shifts belong to Bowen and Lanford [1]; the maximal-entropy Markov law is the Parry construction [3]. Our residual claim is only the explicit symbolic-dynamical synthesis, the sharp odd-modulus correlation rate in this notation, and the two-statistic rigidity statement. No absolute novelty or priority claim is made.

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